

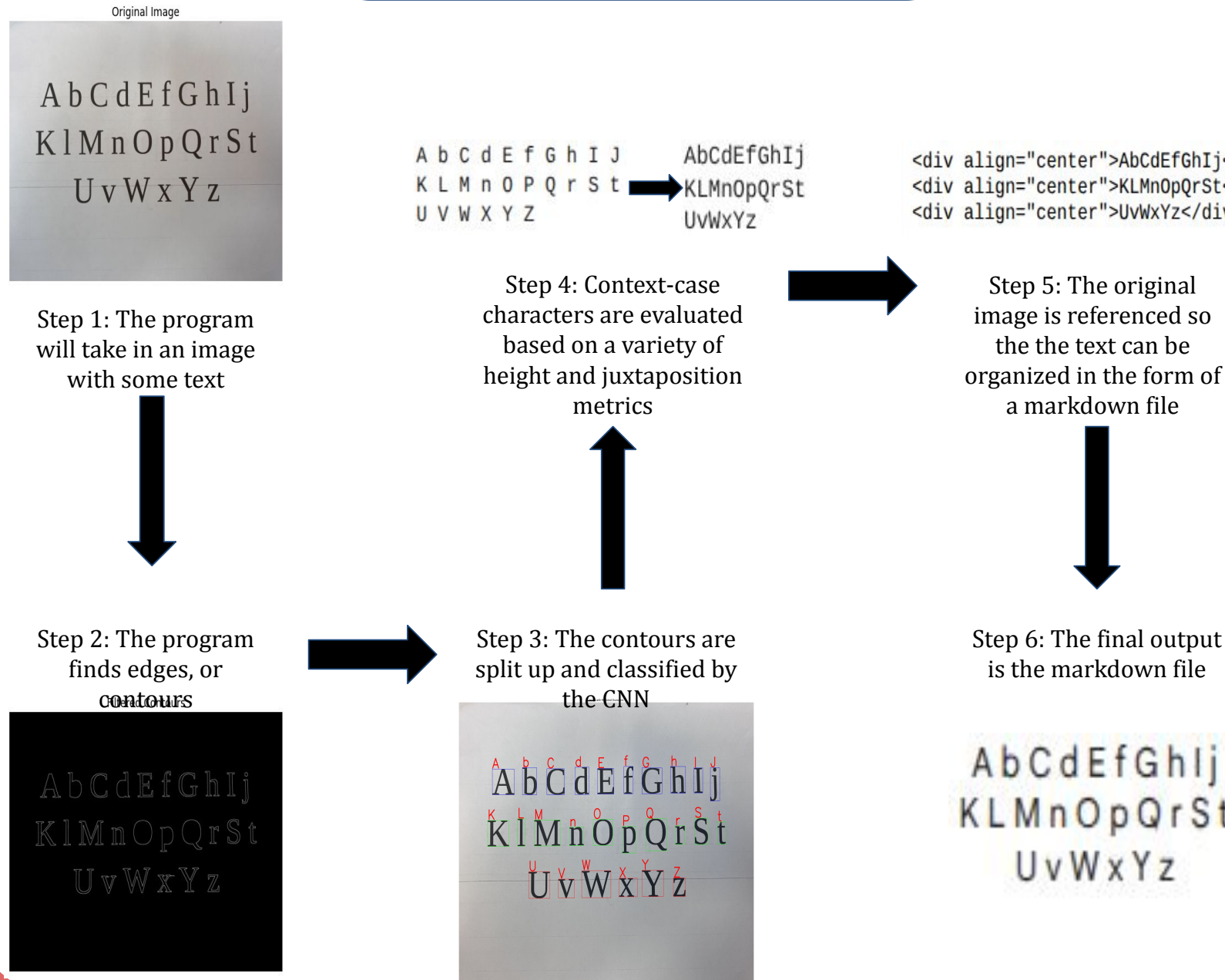
Introduction

Optical Character Recognition (OCR) is the process of converting text in images into digital, editable text. OCR has many real-world applications, including document digitization, handwriting recognition, and accessibility tools. These features are incredibly important in the development of technology, as there is a lot of digitization in the modern day and the incorporation of more data and media into technologies such as AI can help to fast-track their development. This project focuses on developing a convolutional neural network (CNN) capable of recognizing handwritten and printed characters using the EMNIST ByMerge dataset. The final system processes an image containing text, identifies individual characters, reconstructs the detected text, and exports the results into a Markdown-formatted file.

Novelty

Unlike many basic OCR demonstrations that only classify isolated characters, this project implements a full OCR pipeline that reconstructs complete text from an image. The system combines image segmentation, CNN-based character recognition, contextual reconstruction, and Markdown export functionality into a single workflow. The custom context function adds another layer of intelligence by using bounding boxes and character juxtaposition to resolve ambiguous predictions between uppercase and lowercase letters. This allows the model output to be converted into a structured and editable document format rather than simply displaying predicted characters individually.

Process



Creating an Optical Character Recognition Model Using the EMNIST Dataset

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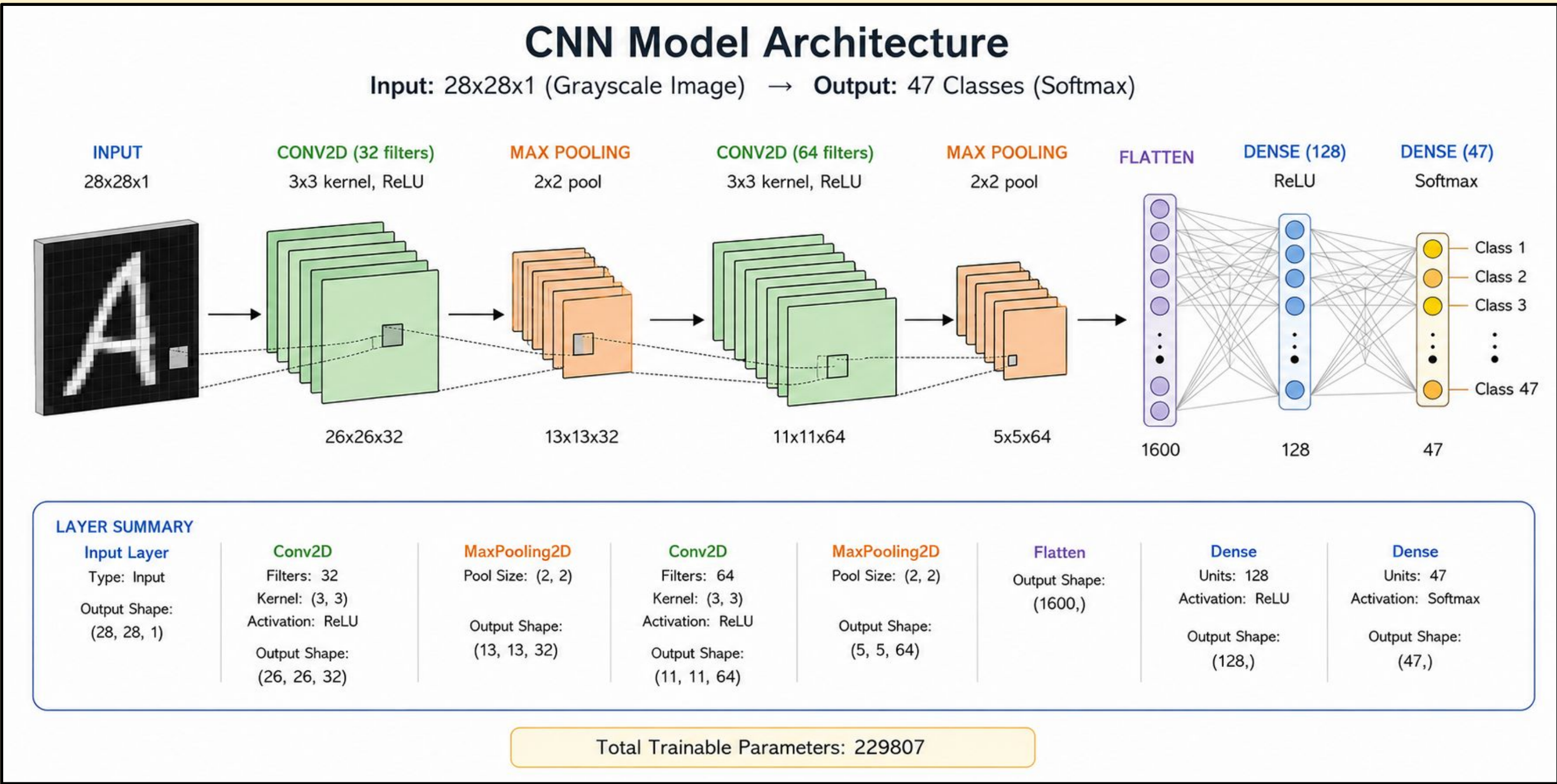


Fig. 1: CNN Model Architecture

Methods

The core of this OCR system is a sequential Convolutional Neural Network (CNN) designed to handle the 47-class complexity of the EMNIST ByMerge dataset. The input layer accepts 28x28x1 grayscale images. The architecture begins with a 2D convolutional layer utilizing 32 filters and a 3x3 kernel with ReLU activation to extract primary spatial features. This is followed by a 2x2 Max Pooling layer to reduce dimensionality while preserving essential features. A second convolutional layer with 64 filters and a 3x3 kernel further refines feature extraction, followed by another 2x2 Max Pooling layer. The multi-dimensional output is then flattened into a 1,600-node vector and passed through a dense layer of 128 neurons with ReLU activation. The final output layer consists of 47 neurons with a Softmax activation function, corresponding to the total number of character categories in the ByMerge labels. In addition, unlike standard classifiers, this project implements a comprehensive end-to-end pipeline. The workflow begins with image segmentation to isolate individual characters, which are then fed into the trained CNN for prediction. To address the inherent ambiguity between certain uppercase and lowercase characters, a custom context function was developed. This function utilizes bounding box coordinates and character scale heuristics to differentiate between similar shapes (e.g., 'o' vs 'O') based on their relative size and positioning within a line. The final stage involves text reconstruction where the predicted characters are organized and exported into a Markdown-formatted document, preserving alignment and line breaks.

Results

The model was trained on the EMNIST ByMerge training set, which contains 697,932 images, and validated against a test set of 116,323 images. The network utilized the Adam optimizer and categorical cross-entropy as the loss function. Training was conducted over 3 epochs with a batch size of 10,000. The final architecture consists of 229,807 trainable parameters. During evaluation, the CNN achieved high classification accuracy across the 47 character classes, successfully identifying both standard printed fonts and varied handwriting styles. Beyond raw classification accuracy, the integration of the contextual analysis function significantly improved the readability of the reconstructed text. By evaluating the scale of characters within their specific bounding boxes, the system more accurately assigned cases (upper vs. lower) which are often indistinguishable in isolation. Test results demonstrated that the pipeline could transform complex multi-line image-based text into editable digital formats while maintaining the original structural intent. The final output successfully generated Markdown files (e.g., predicted_text_aligned_output.md) that mirrored the alignment and sequence of the input handwritten data. While the model performed exceptionally well on clearly segmented characters, minor errors persisted in cases of highly distorted handwriting, suggesting areas for future refinement.

Conclusion

This project demonstrates how convolutional neural networks can be applied to OCR tasks involving handwritten and printed text recognition. By combining character segmentation, CNN classification, contextual analysis, and text reconstruction, the system was able to transform image-based text into editable Markdown documents. The addition of the context function improved the handling of visually similar characters and increased overall reconstruction accuracy. Future improvements could include stronger handling of difficult handwriting, expanded formatting preservation, and support for multi-line document structure reconstruction.



Link to Project Demo

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Systems Research Lab
2026

